Klint Kanopka, Ph.D.

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Academic Appointments

2023 – .... **Assistant Professor of Applied Statistics**, New York University, Steinhardt School of Culture, Education, and Human Development, Department of Applied Statistics, Social Science, and the Humanities

Education

2017 – 2023 **Ph.D., Stanford University** Graduate School of Education
Dissertation title: *Computational Validity.*

2019 – 2021 **M.S., Stanford University** Computer Science

2010 – 2012 **M.S., Drexel University** Education

2001 – 2005 **B.S., Drexel University** Physics

Publications

Journal Articles

1. Lu, Y.†, Harel, D., **Kanopka, K.**, Carrier, M.-E., Kwakkenbos, L., Bartlett, S. J., Fortuné, C., Gietzen, A., Guillot, G., Amanda, L.-J., Malcarne, V. L., Mayes, M. D., Richard, M., Sauvé, M., Stempel, J., Mouthon, L., Benedetti, A., & Thombs, B. D. (2026). Comparing novel and legacy health assessment questionnaire scoring methods: A scleroderma patient-centered intervention network cohort study. *Journal of Rheumatology*. `https://doi.org/10.3899/jrheum.2025-1069`

2. Ramamurthy, M., **Kanopka, K.**, Siebert, J. M., Yan, L., Townley-Flores, C., Zegers, M., Pei, F., Bell, P., Catts, H., Gorno-Tempini, M. L., & Yeatman, J. D. (2026). Visual processing is a language-agnostic window into heterogeneity in early reading development. *Current Biology*. `https://doi.org/10.1016/j.cub.2026.04.055`

3. Zhang, L., Rahal, C., **Kanopka, K.**, Ulitzsch, E., Zhang, Z., & Domingue, B. W. (2026). Evaluating model predictive performance in confirmatory factor analysis with binary outcomes using the intermodel vigorish. *Multivariate Behavioral Research*, 1–20. `https://doi.org/10.1080/00273171.2026.2645212`

4. Domingue, B. W., Braginsky, M., Caffrey-Maffei, L., Gilbert, J. B., **Kanopka, K.**, Kapoor, R., Lee, H., Liu, Y., Nadela, S., Pan, G., et al. (2025). An introduction to the item response warehouse (irw): A resource for enhancing data usage in psychometrics. *Behavior Research Methods*, *57*(10), 1–11. `https://doi.org/10.3758/s13428-025-02796-y`

5. Domingue, B. W., **Kanopka, K.**, Ulitzsch, E., & Zhang, L. (2025). Implied probabilities of polytomous response functions for model-based prediction and comparison. *Behaviormetrika*, 1–23. `https://doi.org/10.1007/s41237-025-00262-9`

6. Domingue, B. W., Rahal, C., Faul, J., Freese, J., **Kanopka, K.**, Rigos, A., Stenhaug, B., & Tripathi, A. S. (2025). The intermodel vigorish (imv) as a flexible and portable approach for quantifying predictive accuracy with binary outcomes. *PloS one*, *20*(3), e0316491. `https://doi.org/10.1371/journal.pone.0316491`

7. **Kanopka, K.**, & Domingue, B. W. (2025). A position-sensitive mixture item response model. *Journal of Educational and Behavioral Statistics*, *50*(6), 1026–1059. `https://doi.org/10.3102/10769986241289399`

† denotes student collaborators

8. Kroll, E., Hessdorf, S., Brandis, A., **Kanopka, K.**, Kohlmeier, J., Heard, H., Zant, I., Stolzenbach, S., & Fenkel, C. (2025). Examining caregiver-adolescent symptom agreement and its influence on treatment outcomes: A quantitative analysis. *Frontiers in Psychology*, *16*, 1676629.

9. Ma, W. A., Richie-Halford, A., Burkhardt, A. K., **Kanopka, K.**, Chou, C., Domingue, B. W., & Yeatman, J. D. (2025). Roar-cat: Rapid online assessment of reading ability with computerized adaptive testing. *Behavior Research Methods*, *57*(1), 1–17. https://doi.org/10.3758/s13428-024-02578-y

10. Ramamurthy, M., **Kanopka, K.**, Richie-Halford, A., Domingue, B. W., Pei, F., Bell, P., Yan, L., Hartsough, A., Luisa, G. T. M., & Yeatman, J. D. (2025). Design and validation of a rapid visual processing measure for screening reading difficulties in early childhood. *Behavior Research Methods*, *57*. https://doi.org/10.3758/s13428-025-02739-7

11. Caffarra, S., **Kanopka, K.**, Kruper, J., Richie-Halford, A., Roy, E., Rokem, A., & Yeatman, J. D. (2024). Development of the alpha rhythm is linked to visual white matter pathways and visual detection performance. *Journal of Neuroscience*, *44*(6). https://doi.org/10.1523/JNEUROSCI.0684-23.2023

12. Domingue, B. W., **Kanopka, K.**, Kapoor, R., Pohl, S., Chalmers, R. P., Rahal, C., & Rhemtulla, M. (2024). The intermodel vigorish as a lens for understanding (and quantifying) the value of item response models for dichotomously coded items. *psychometrika*, *89*(3), 1034–1054. https://doi.org/10.1007/s11336-024-09977-2

13. **Kanopka, K.**, Claro, S., Loeb, S., West, M., & Fricke, H. (2024). Are changes in reported social-emotional skills just noise? the predictive power of longitudinal differences in self-reports. *AERA Open*, *10*, 23328584241233277. https://doi.org/10.1177/23328584241233277

14. Kapoor, R., Fahle, E., **Kanopka, K.**, Klinowski, D., Ribeiro, A. T., & Domingue, B. W. (2024). Differences in time usage as a competing hypothesis for observed group differences in accuracy with an application to observed gender differences in pisa data. *Journal of Educational Measurement*, *61*(4), 682–709. https://doi.org/10.1111/jedm.12419

15. Moeller, K. J., **Kanopka, K.**, French, J., Hook, T., & Sedighi, M. (2024). Educational capitalisation: A co-formational feminist framework for conceptualising investment in for-profit education within the racialised and gendered political economy. *Globalisation, Societies and Education*, 1–15. https://doi.org/10.1080/14767724.2024.2397811

16. Trejo, S., & **Kanopka, K.** (2024). Using the phenotype differences model to identify genetic effects in samples of partially genotyped sibling pairs. *Proceedings of the National Academy of Sciences*, *121*(49), e2405725121. https://doi.org/10.1073/pnas.2405725121

17. Biernacki, P. J., Altavilla, J., **Kanopka, K.**, Hsieh, H., & Solano-Flores, G. (2023). Long-term english learners' mathematics course trajectories: Downstream consequences of early remediation on college preparation. *International Multilingual Research Journal*, *17*(2), 122–138. https://doi.org/10.1080/19313152.2022.2137910

18. Ramamurthy, M., Richie-Halford, A., **Kanopka, K.**, Hartsough, A., Osuna, K., Gorno-Tempini, M. L., & Yeatman, J. D. (2023). Can a gamified, rapid, online assessment of letter encoding ability in kindergarten and first grade children predict future reading development? *Journal of Vision*, *23*(9), 5858–5858. https://doi.org/10.1167/jov.23.9.5858

19. Ulitzsch, E., Domingue, B. W., Kapoor, R., **Kanopka, K.**, & Rios, J. A. (2023). A probabilistic filtering approach to non-effortful responding. *Educational Measurement: Issues and Practice*, *42*(3), 50–64. https://doi.org/10.1111/emip.12567

20. Domingue, B. W., **Kanopka, K.**, Mallard, T. T., Trejo, S., & Tucker-Drob, E. M. (2022). Modeling interaction and dispersion effects in the analysis of gene-by-environment interaction. *Behavior genetics*, *52*(1), 56–64. https://doi.org/10.1007/s10519-021-10090-8

21. Domingue, B. W., **Kanopka, K.**, Stenhaug, B., Sulik, M. J., Beverly, T., Brinkhuis, M., Circi, R., Faul, J., Liao, D., McCandliss, B., et al. (2022). Speed–accuracy trade-off? not so fast: Marginal changes

in speed have inconsistent relationships with accuracy in real-world settings. *Journal of Educational and Behavioral Statistics*, *47*(5), 576–602. https://doi.org/10.3102/10769986221099906

22. Domingue, B. W., **Kanopka, K.**, Trejo, S., Rhemtulla, M., & Tucker-Drob, E. M. (2022). Ubiquitous bias and false discovery due to model misspecification in analysis of statistical interactions: The role of the outcome's distribution and metric properties. *Psychological methods*. https://doi.org/10.1037/met0000532

23. Domingue, B. W., **Kanopka, K.**, Stenhaug, B., Soland, J., Kuhfeld, M., Wise, S., & Piech, C. (2021). Variation in respondent speed and its implications: Evidence from an adaptive testing scenario. *Journal of Educational Measurement*, *58*(3), 335–363. https://doi.org/10.1111/jedm.12291

24. Yeatman, J. D., Tang, K. A., Donnelly, P. M., Yablonski, M., Ramamurthy, M., Karipidis, I. I., Caffarra, S., Takada, M. E., **Kanopka, K.**, Ben-Shachar, M., et al. (2021). Rapid online assessment of reading ability. *Scientific reports*, *11*(1), 6396. https://doi.org/10.1038/s41598-021-85907-x

Book Chapters

1. Ruiz-Primo, M. A., **Kanopka, K.**, & Hernandez, P. A. (2025). Measuring, assessing, or just testing student achievement? evidence from some states' testing programs in the united states. In L. Cohen-Vogel, P. Youngs, & J. Scott (Eds.), *Handbook of education policy research* (2nd, pp. 511–542). American Educational Research Association. https://doi.org/10.2307/jj.31510356.29

Conference Proceedings

1. Mongkhonvanit, K., **Kanopka, K.**, & Lang, D. (2019). Deep knowledge tracing and engagement with moocs. *Proceedings of the 9th international conference on learning analytics & knowledge*, 340–342. https://doi.org/10.1145/3303772.3303830

Preprints

1. **Kanopka, K.**, Zhang, X.†, & Harel, D. (2026). A markov chain monte carlo procedure with simulated annealing for shortening assessments. *PsyArXiv*. https://doi.org/10.31234/osf.io/mwq7u_v1

2. Pizza, J.†, & **Kanopka, K.** (2026). Comparing campaign rhetoric and legislative behavior through issue-specific polarization in the us house of representatives, 2021–2025. *SocArXiv*. https://doi.org/10.31235/osf.io/sr7jq_v1

3. Shen, R.†, & **Kanopka, K.** (2026b). Understanding the evolution of individual response process: An exploratory approach. *PsyArXiv*. https://doi.org/10.31234/osf.io/yczgu_v1

4. Student, S., & **Kanopka, K.** (2026b). Making room for theory in dif and measurement invariance analysis with covariate effects on groups of items. *PsyArXiv*. https://doi.org/10.31234/osf.io/98y6k_v1

5. Van Loon, A., & **Kanopka, K.** (2026). Using large language models as a source of human behavioral data in social science experiments. *SocArXiv*. https://doi.org/10.31235/osf.io/y74mu_v1

6. Alvero, A., Dong, R., **Kanopka, K.**, & Lang, D. (2025). Algorithmic tradeoffs, applied nlp, and the state-of-the-art fallacy. *arXiv*. https://doi.org/10.48550/arXiv.2509.08199

7. Ma, W. A., Liu, Y., **Kanopka, K.**, Ma, W., & Domingue, B. W. (2025). A comparison of the predictive performance of continuous and class-based latent trait models. *PsyArXiv*. https://doi.org/10.31234/osf.io/acwdz_v1

8. Zhang, L., Liu, Y., Molenaar, D., Gilbert, J., **Kanopka, K.**, & Domingue, B. W. (2025). Realistic simulation of item difficulties. *PsyArXiv*. https://doi.org/10.31234/osf.io/jbhxy_v3

9. Deng, S.†, Siswandjo, J.†, & **Kanopka, K.** (2024). Mapping the participation of venture capital in education: An investment network perspective. *SocArXiv*. https://doi.org/10.31235/osf.io/pekyg

Software Packages

1. **Kanopka, K.** (2026b). *Mixedsubjectsirt: Data augmentation for psychometric modeling using the mixed-subjects design* [R package version 0.1.0]. http://klintkanopka.com/mixedsubjectsirt/
2. van Loon, A., & **Kanopka, K.** (2026). *Mixedsubjects: Conducting social science experiments using the mixed-subjects design* [R package version 1.0.0]. http://klintkanopka.com/mixedsubjects/
3. **Kanopka, K.**, & Deng, S.[†]. (2025). *Meow: Unified framework for computer adaptive testing simulations* [R package version 1.0.0]. http://klintkanopka.com/meow/

Policy Briefs

1. **Kanopka, K.**, Claro, S., Loeb, S., West, M. R., & Fricke, H. (2020). What do changes in social-emotional learning tell us about changes in academic and behavioral outcomes? *Policy Analysis for California Education*. https://eric.ed.gov/?id=ED609206

Presentations

Invited Talks

1. **Kanopka, K.** Large language models as potentially informative respondents: Data augmentation for psychometric applications. In: University of California, Los Angeles Quantitative Pscyhology Brown Bag. 2026.
2. **Kanopka, K.** Reproducing expert judgement with shortened surveys using simulated annealing. In: Metrics, Models: A Seminar Series for Health, and Social Data Science at the University of Oxford. 2025.
3. **Kanopka, K.** A position-sensitive mixture item response model. In: Fordham Psychometrics Seminar at Fordham University. 2024.
4. **Kanopka, K.**, & Moeller, K. Following the capital: Using investment networks to understand venture capital's influence on education. In: Society for Research into Higher Education Digital University Network Event on the Political Economy of EdTech. 2023.
5. **Kanopka, K.** Good luck with the blue book! an irt mixture model for position effects. In: Berkeley Evaluation & Assessment Research (BEAR) Seminar at UC Berkeley. 2021.

Conference Presentations

1. Shen, R.[†], & **Kanopka, K.** A hidden markov approach to understand behavior transitions in test-taking processes. In: Annual Meeting of the National Council on Measurement in Education. 2026.
2. Student, S., & **Kanopka, K.** Decomposing dif into item group and item-specific effects. In: Modern Modeling Methods. 2026.
3. Xiao, X., Domingue, B. W., Frank, M., & **Kanopka, K.** Good kitty, bad bank? rescoring miscalibrated cats improves accuracy. In: Annual Meeting of the National Council on Measurement in Education. 2026.
4. **Kanopka, K.** Prediction quality of a range of polytomous models applied to a large data corpus. In: Annual Meeting of the National Council on Measurement in Education. 2025.
5. **Kanopka, K.**, Deng, S.[†], & Lu, Y.[†]. Item pool maintenance in computer adaptive tests: A network approach. In: International Meeting of the Psychometric Society. 2025.
6. **Kanopka, K.**, van Loon, A., Huang, Y.[†], Shen, R.[†], & Tang, H.[†]. Data augmentation for psychometric applications using a mixed subjects design. In: Pacific Quantitative Psychometrics Conference. 2025.

7. **Kanopka, K.**, Zhang, X.†, & Harel, D. A markov chain monte carlo procedure with simulated annealing for short form creation. In: International Objective Measurement Workshop. 2025.

8. Lu, Y.†, Huang, Y.†, & **Kanopka, K.** Assessing the representativeness of llm-generated item responses using latent class analysis. In: Annual Meeting of the National Council on Measurement in Education. 2025.

9. Shen, R.†, & **Kanopka, K.** Understanding the evolution of individual response process: An exploratory approach. In: International Meeting of the Psychometric Society. 2025.

10. Domingue, B. W., **Kanopka, K.**, Braginksy, M., Zhang, L., Caffrey-Maffei, L., Kapoor, R., Liu, Y., Zhang, S., & Frank, M. The item response warehouse (irw): A large open repository of response data. In: Annual Meeting of the National Council on Measurement in Education. 2024.

11. Domingue, B. W., **Kanopka, K.**, Caffrey-Maffei, L., Kapoor, R., Zhang, S., & Liu, Y. The item response warehouse (irw): A large open repository of response data. In: Annual Meeting of the National Council on Measurement in Education. 2024.

12. Domingue, B. W., **Kanopka, K.**, Ulitzsch, E., & Zhang, L. Implied probabilities of polytomous response functions for model-based prediction and comparison. In: Annual Meeting of the National Council on Measurement in Education. 2024.

13. **Kanopka, K.**, & Harel, D. A lasso approach for short form item selection. In: International Meeting of the Psychometric Society. 2024.

14. Domingue, B. W., **Kanopka, K.**, Kapoor, R., Pohl, S., Chalmers, R. P., Rahal, C., & Rhemtulla, M. Applying the intermodel vigorish to quantify the value of item response modeling. In: Annual Meeting of the National Council on Measurement in Education. 2023.

15. **Kanopka, K.**, & Domingue, B. W. Projecting the performance of polytomous item response models onto a common scale with the intermodel vigorish. In: International Conference on Computational and Methodological Statistics. 2023.

16. **Kanopka, K.**, Domingue, B. W., Ulitzsch, E., Kapoor, R., Pohl, S., Chalmer, R. P., Rahal, C., & Rhemtulla, M. Bookmaking for item responses: Prediction, profits, and the imv. In: International Objective Measurement Workshop. 2023.

17. **Kanopka, K.**, & Hook, T. J. Following the capital: Using investment networks to understand venture capital's influence on education. In: Annual Meeting of the Comparative and International Education Society. 2023.

18. **Kanopka, K.**, Lang, D., & Alvero, A. Weighing algorithmic tradeoffs: Observations from 60,000 admission essays. In: Annual Meeting of the National Council on Measurement in Education. 2023.

19. **Kanopka, K.**, Yeatman, J., & Burkhardt, A. Incorporating response time using drift diffusion models in an online reading assessment. In: Annual Meeting of the National Council on Measurement in Education. 2023.

20. Kapoor, R., **Kanopka, K.**, & Domingue, B. W. Revisiting angrist et al. (2021): Merging international assessment data is not easy. In: Annual Meeting of the National Council on Measurement in Education. 2023.

21. Kapoor, R., Ruiz-Primo, M. A., Hernandez, P., & **Kanopka, K.** Online response process procedures: Affordances and constraints to collect, analyze, and interpret validity information. In: Annual Meeting of the American Educational Research Association. 2023.

22. Ulitzsch, E., Rios, J. A., Kapoor, R., **Kanopka, K.**, & Domingue, B. W. A probabilistic filtering approach to accounting for noneffortful responding. In: Annual Meeting of the National Council on Measurement in Education. 2023.

23. Wang, N., Li, M., **Kanopka, K.**, Dong, D., Hernandez, P., & Ruiz-Primo, M. A. Modeling context characteristics for contextualized assessment: A bayesian contextualized item response model. In: Annual Meeting of the National Council on Measurement in Education. 2023.

24. **Kanopka, K.** Profiling changes in student writing: From failure to success. In: Annual Meeting of the National Council on Measurement in Education. 2022.

25. **Kanopka, K.**, & Domingue, B. W. Good luck with the blue book: Computational approaches to fairness and validity in admissions testing. In: Annual Meeting of the American Educational Research Association. 2022.

26. Moeller, K., & **Kanopka, K.** Who profits from edtech? an intersectional feminist analysis of venture capital investment in education. In: Annual Meeting of the American Educational Research Association. 2022.

27. **Kanopka, K.**, & Domingue, B. Addressing the blue book problem: Application of a position-sensitive irt mixture model to enem 2014. In: International Meeting of the Psychometric Society. 2021.

28. **Kanopka, K.**, & Domingue, B. W. An irt mixture model for item position effects. In: Annual Meeting of the National Council on Measurement in Education. 2021.

29. **Kanopka, K.**, Hernandez, P., Wang, N., Ruiz-Primo, M. A., Li, M., & Minstrell, J. An application of multiple correspondence analysis to the experimental study of item context. In: Annual Meeting of the American Educational Research Association. 2021.

30. Moeller, K., **Kanopka, K.**, & French, J. Venture capitalists as educational actors: Understanding the racialized political economy of silicon valley investments in education technology. In: Annual Meeting of the American Educational Research Association. 2021.

31. Moeller, K., **Kanopka, K.**, & French, J. Venture capitalists as educational actors: Understanding the racialized political economy of silicon valley investments in education technology. In: Annual Meeting of the Comparative and International Education Society. 2021.

32. Ruiz-Primo, M. A., Li, M., **Kanopka, K.**, Hernandez, P., Dong, D., Wang, N., & Minstrell, J. Analysis and review of context in science: Effects in student performance. In: Annual Meeting of the American Educational Research Association. 2021.

33. Alvero, A., Sedlacek, Q., & **Kanopka, K.** A call for critical and constructive data science in teacher eduaction. In: AERA Conference on Educational Data Science. 2020.

34. Claro, S., **Kanopka, K.**, West, M., Loeb, S., & Fricke, H. Exploring the relationship between changes in social-emotional skills and achievement. In: Annual Meeting of the American Educational Research Association (Conference Cancelled). 2020.

35. Ruiz-Primo, M. A., Li, M., Minstrell, J., Dong, D., **Kanopka, K.**, & Hernandez, P. Mapping the characteristics of contexts in science items: The case of forces and motion items. In: Annual Meeting of the American Educational Research Association (Conference Cancelled). 2020.

36. Hernandez, P., Ruiz-Primo, M. A., Zhai, X., & **Kanopka, K.** Validity study of linked items to determine student fundamental ideas. In: Annual Meeting of the American Educational Research Association. 2019.

37. **Kanopka, K.** Addressing defensive objections: Adversarial examples in automatic essay scoring. In: Annual Meeting of the American Educational Research Association. 2019.

38. **Kanopka, K.** Diagnosing non-parallelism in hierarchically contextualized physics assessments. In: Annual Meeting of the National Association for Research in Science Teaching. 2019.

39. Munoz-Najar Galvez, S., **Kanopka, K.**, & Alvero, A. Identifying (dis)continuities in edtech's discourse of invention. In: Text Analysis Across Domains. 2019.

40. Ruiz-Primo, M. A., Li, M., Minstrell, J., Zhai, X., Dong, D., **Kanopka, K.**, & Hernandez, P. Testing the generalization to the domain inference: The use of contextualized clusters of items. In: Annual Meeting of the National Council on Measurement in Education. 2019.

41. Ruiz-Primo, M. A., Li, M., Minstrell, J., Zhai, X., **Kanopka, K.**, Hernandez, P., & Dong, D. Contextualized science assessments: Addressing the use of information and generalization of inferences of students' performance. In: Annual Meeting of the American Educational Research Association. 2019.

42. Zhai, X., Ruiz-Primo, M. A., Li, M., Dong, D., **Kanopka, K.**, Hernandez, P., & Minstrell, J. Using many-facet rasch model to examine student performance on contextualized science assessment. In: Annual Meeting of the American Educational Research Association. 2019.

Awards and Fellowships

2024 **W. Gabriel Carras Research Award**, New York University Steinhardt School of Culture, Education, and Human Development

Brenda H. Loyd Outstanding Dissertation Award, National Council on Measurement in Education

2021 **Gerald J Lieberman Fellowship**, Stanford University

Summer Internship, Educational Testing Service

2017 **Stanford Graduate Fellowship in Science & Engineering**, Stanford University

Courses Taught

New York University

2024 – · · · · **APSTA-GE 2094 (APSY-GE 2524)**: Modern Approaches in Measurement

APSTA-GE 2044: Generalized Linear Models and Extensions

2023 – · · · · **APSTA-GE 2352**: Practicum in Applied Statistics: Statistical Computing

Stanford University

2023 **EDUC 423B (SOC 302B)**: Introduction to Education Data Science: Data Analysis

2019 **EDUC 200A**: Introduction to Data Analysis and Interpretation

Professional Activities

Professional Memberships

2017 – · · · · American Educational Research Association

2018 – · · · · National Council on Measurement in Education

2019 – · · · · Psychometric Society

2023 – · · · · Comparative and International Education Society

Service to Field

Editorial New Directions for Evaluation (Editorial Board)

Reviewer AERA Open

Behavior Genetics

Professional Activities (continued)

British Journal of Mathematical and Statistical Psychology
Educational Assessment
Educational Evaluation and Policy Analysis
Educational Measurement: Issues and Practice
Educational Researcher
Journal of Leisure Research
Journal of Medical Internet Research: Pediatrics and Parenting
New Directions for Evaluation
Proceedings of the National Academy of Sciences

Organized Sessions

2023 **CMStatistics**: Applied statistical and psychometrics issues in measurement (with Daphna Harel)